RoboCore Zambia

GET STARTED HERE

www.plastalbotbuilders.com

On this page you will find:

- Assembly and first-run guide
- Step-by-step classroom lessons
- Programming environments: blocks, Python, C++
- Firmware, example code and libraries
- 3D-print files for the chassis and the arm
- Support forum and teacher resources

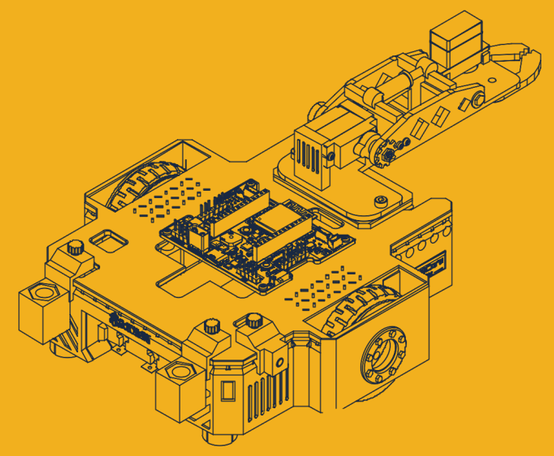
BEFORE YOU START

- 3 x 18650 lithium cells (not included) and a USB-C charger
- A computer or tablet with a browser, or the Arduino IDE
- Charge the robot fully before the first lesson
- Fit the arm module only after the drive base runs

Replacement parts, add-on sensors and the arm module:
plastalbotbuilders.com/robocore

Design files are published. Print it, fix it, improve it, share it.

SCAN FOR DOCS, CODE AND LESSO



PLASTAL-BOT BUILDERS

Plastal-Bot Builders is a Lusaka makerspace that teaches robotics, coding and fabrication to learners and teachers across Zambia. RoboCore is our own controller family, designed so a class can build, program and repair its robots locally.

JOIN THE ROBOCORE COMMUNITY

Share what your class builds, find lesson plans and compare notes with other schools. Replacement parts, add-on sensors and the arm module are available from us; the design files are published so you can also print and fix parts yourself.

RoboCore V2 box, inside, tile 1 of 6 (row 1, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

ASTAL-BOT
UILDERS

AND LESSONS

RoboCore V2 box, inside, tile 2 of 6 (row 1, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.

WORKSHOPS · TEACHER TRAINING · ROBOTICS CLUBS · FABRICATION

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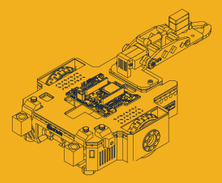
SCAN

LESSON PLANS · FORUM · SPARE PARTS · DESIGN FILES

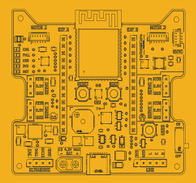
www.plastalbotbuilders.com/community

THIS KIT INCLUDES

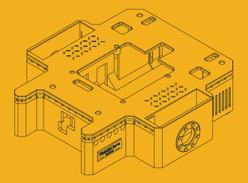
RoboCore V2 robot, assembled



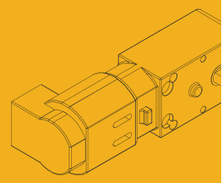
RoboCore V2 controller (on the robot)



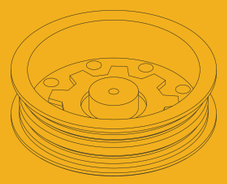
3D-printed chassis (on the robot)



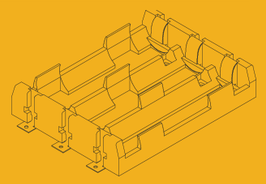
Encoder motors DG01D-E (2, on the



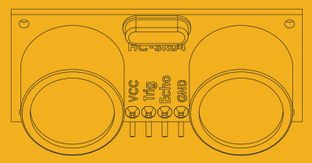
Wheels 56 mm, TPU tyres (2)



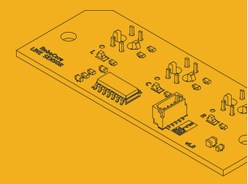
Battery holder, 3 x 18650 (on the robot)



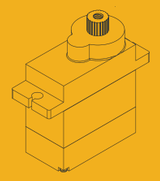
Ultrasonic sensor HC-SR04 (on the robot)



Line sensor, 3 channels (on the robo



Micro servos (2)



Arm boom



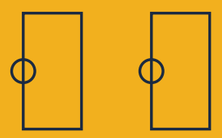
Arm base



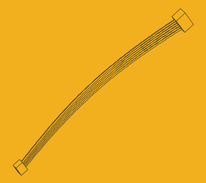
Claw jaws (2)



Ballast clips (2)



Motor encoder cables (2)



USB-C cable



Screws, nuts and quick-start card



For replacement parts and the arm module visit plastalbotbuilders.com/robocore

ROBOCORE V2 · BUILD. CODE. CREATE.

Designed and built in Zambia by Plastal-Bot Builders, Lusaka

RoboCore V2 box, inside, tile 3 of 6 (row 2, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.

SCAN

2, on the robot)



the robot)





rt card



RoboCore V2 box, inside, tile 4 of 6 (row 2, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.



RoboCore V2 box, inside, tile 5 of 6 (row 3, column 1), 1:1. Overlap 10 mm with the neighbouring tiles.



RoboCore V2 box, inside, tile 6 of 6 (row 3, column 2), 1:1. Overlap 10 mm with the neighbouring tiles.